SMART SCHOLARSHIP DISTRIBUTION PLATFORM

The Smart Scholarship Distribution Platform is a centralized web-based system that connects Students, Admin, Colleges, and Sponsors in a secure and intelligent scholarship management process. The system ensures fairness through AI-based eligibility prediction and multi-level verification before fund disbursement and certificate generation.

🔷 1️⃣ ADMIN MODULE

🔹 Overview

The Admin module is the core control unit of the system. The Admin manages scholarship creation, verifies student applications, performs AI-based eligibility prediction, sends verification requests to colleges, gives final approval or rejection, monitors fund disbursement, generates certificates, and analyzes reports and feedback.

🔹 Detailed Working Flow

1. Login & Authentication

The Admin logs into the system using secure credentials.
If authentication fails, the system redirects to the login page.
After successful login, the Admin dashboard is displayed.

2. Manage Scholarship

Admin can create and manage scholarship programs.

During scholarship creation, the Admin defines:

- Scholarship Name
- Scholarship Description
- Eligibility Criteria (e.g., minimum marks, income limit)
- Minimum Required Eligibility Score (e.g., Score ≥ 70 after AI prediction)
- Payment Amount per student
- Maximum Number of Students allowed

Admin can also update, deactivate, or delete scholarships.

3. Initial Student Application Verification

When a student applies:

- Admin reviews uploaded documents
- Checks completeness of application
- Validates basic eligibility conditions

Applications that do not meet basic criteria are rejected immediately.

4. AI-Based Eligibility Prediction

For valid applications:

- The system calculates an eligibility score using the prediction model
- The predicted score is compared with the required minimum score defined in the scholarship

If Predicted Score < Required Score → Application Rejected
If Predicted Score ≥ Required Score → Proceed to College Verification

This step reduces unnecessary workload on colleges.

5. Send Verification Request to College

Admin forwards only predicted-eligible applications to the respective college for academic verification.

6. Receive College Verification Result

College verifies academic authenticity and sends response.

If College Confirms → Admin gives Final Approval
If College Rejects → Admin Rejects Application

Admin has final decision authority.

7. Fund Disbursement Monitoring

After final approval:

- Sponsor releases funds
- Admin monitors transaction
- Updates payment confirmation status

8. Certification Module

After payment confirmation:

- Admin generates digital scholarship certificate
- Certificate becomes available for student download
- Certificate records are stored securely

9. Feedback

Admin can:

- View feedback from students

10. Generate Reports:

- Scholarship-wise student list
- Approved & rejected students

- Fund utilization report
- Feedback analytics report

11. Logout

Admin logs out securely, ending the session.

🔷 2️⃣ STUDENT MODULE

🔹 Overview

The Student module enables students to explore available scholarships, apply by submitting required documents, track the multi-level verification process, receive scholarship funds upon approval, download digital certificates, and provide feedback.

The student interacts with the system from application submission to final certificate download, but the eligibility prediction and approval decisions are controlled by the Admin to ensure fairness and transparency.

1. Registration & Login

The student registers in the system by providing personal and academic details.
After successful registration, the student logs in using valid credentials.

If authentication fails, the system redirects to the login page.
After successful login, the student dashboard is displayed.

2. View Available Scholarships

The student can browse scholarships created by Admin or Sponsor.
For each scholarship, the student can view:

• Scholarship Name
• Scholarship Description
• Eligibility Criteria (e.g., minimum marks, income limit)
• Minimum Required Eligibility Score (e.g., Score ≥ 70 after AI prediction)
• Payment Amount per student
• Maximum Number of Students allowed

This allows the student to determine whether they meet the requirements before applying.

3. Apply for Scholarship

If eligible, the student proceeds to apply:

• Fill out the scholarship application form
• Upload academic documents (mark sheets, certificates)
• Upload income proof and identity proof
• Submit the application

After submission, the application status becomes:
“Under Admin Verification.”

4. Application Status Tracking

The student can monitor application progress in real-time.
The system displays different stages such as:

• Under Admin Verification
• Under AI Eligibility Prediction
• Under College Verification
• Approved
• Rejected

This ensures transparency throughout the process.

5. View Final Decision

After Admin completes prediction and college verification:

• If approved → Student receives approval notification
• If rejected → Student receives rejection notification

The final decision is controlled by the Admin.

6. View Payment Status

If approved:

• Student can check fund disbursement status
• View transaction confirmation
• See payment details (amount, date, scholarship name)

7. Download Scholarship Certificate

After payment confirmation:

• The system enables certificate download
• Student can download digital scholarship certificate
• Certificate serves as official proof of scholarship award

8. Submit Feedback

The student can provide feedback regarding:

• Scholarship process
• Verification experience
• Payment process
• Overall system performance

Feedback helps improve system transparency and performance.

9. Logout

After completing activities, the student logs out securely, ending the session.

- **Functional Responsibilities of Student Module**

• Scholarship browsing
• Application submission
• Document upload
• Application status tracking
• Payment monitoring
• Certificate download
• Feedback submission

🔷 3️⃣ SPONSOR MODULE

🔹 Overview

The Sponsor module enables individuals, organizations, or institutions to create and fund scholarship programs. Sponsors can define scholarship criteria, set eligibility score requirements, specify funding limits, view approved students, release funds, monitor disbursement status, track scholarship impact, and provide feedback.

While sponsors can create and manage scholarships, the final approval of students is handled by the Admin to ensure fairness and centralized control.

1. Registration & Login

The sponsor registers by providing organization or personal details.
After registration, the sponsor logs in using valid credentials.

If authentication fails, the system redirects to the login page.
After successful login, the sponsor dashboard is displayed.

2. Manage Scholarship

The sponsor can create and manage scholarship programs.

During scholarship creation, the sponsor defines:

• Scholarship Name
• Scholarship Description
• Eligibility Criteria (e.g., minimum marks, income limit)
• Minimum Required Eligibility Score (e.g., Score ≥ 70 after AI prediction)
• Payment Amount per student
• Maximum Number of Students allowed

The sponsor can also:

• Update scholarship details
• Deactivate or close scholarship
• View active and completed scholarships

This allows sponsors to control funding distribution based on predefined rules.

3. View Approved Students

After Admin completes verification and final approval:

• Sponsor can view the list of approved students
• View student details (name, college, eligibility score, scholarship name)
• Filter students scholarship-wise

Sponsors can only fund students who are officially approved by Admin.

4. Release Scholarship Funds

Sponsor selects approved students and proceeds with fund disbursement:

• Confirm payment amount
• Transfer scholarship funds
• Update transaction status

Once payment is completed, the system updates fund disbursement records.

5. Track Fund Disbursement

The sponsor can monitor:

• Payment history
• Total funded amount
• Scholarship-wise disbursement
• Number of beneficiaries funded

This ensures transparency in financial contribution.

6. View Impact Report

The sponsor can analyze:

• Number of students benefited
• College-wise distribution
• Scholarship performance
• Overall impact statistics

This helps sponsors evaluate the effectiveness of their funding programs.

8. Logout

After completing activities, the sponsor logs out securely, ending the session.

🔹 Functional Responsibilities of Sponsor Module

• Scholarship creation and management
• Define eligibility score requirements
• Define funding limits
• Fund approved students
• Monitor fund disbursement
• View impact analytics

🔷 4️⃣ COLLEGE MODULE

🔹 Overview

The College module plays a crucial role in ensuring the authenticity of student academic records. After the Admin performs AI-based eligibility prediction and the student's predicted score meets the required threshold, the application is forwarded to the respective college for academic verification.

The college verifies the accuracy of academic documents, confirms student enrollment status, and sends verification results back to the Admin for final decision-making. This ensures multi-level validation and prevents fraudulent applications.

1. Registration & Login

The college registers by providing institutional details such as college name, registration ID, and official contact information.

After successful registration, the college logs in using valid credentials.

If authentication fails, the system redirects to the login page.
After successful login, the college dashboard is displayed.

2. Receive Verification Request

After Admin performs eligibility prediction and the student meets the minimum required score:

• The Admin sends a verification request to the respective college
• The college receives notification of pending verification
• The college can view student details and selected scholarship

This ensures that only predicted-eligible students are verified by the college.

3. Verify Academic Records

The college verifies:

• Student marks and grades
• Enrollment status
• Academic year and course details
• Uploaded academic certificates
• Authenticity of documents

This step ensures that the student information provided is genuine and accurate.

4. Validate Scholarship Criteria

The college checks whether:

• The student satisfies academic eligibility criteria
• The student belongs to the institution
• The provided documents match official records

If any mismatch or false information is detected, the application is marked as not verified.

5. Send Verification Result to Admin

After verification, the college sends response:

• Verified → Application forwarded to Admin for final approval
• Not Verified → Application marked for rejection

The Admin retains final decision authority.

6. Track Application Status

The college can monitor:

• Final approval or rejection status
• Fund disbursement updates
• Scholarship allocation details

7. Logout

After completing verification tasks, the college logs out securely, ending the session.

AI/ML Features in Smart Scholarship Distribution Platform

🔷 1️⃣ **Eligibility Score Prediction (Core Feature)**

✅ **What it does:**

Predicts whether a student is eligible for a scholarship based on:

- Academic performance
- Family income
- Category / background
- Attendance
- Extra achievements
- Past academic trend

✅ **Output:**

System generates an **eligibility score (0–100)**.

Example:

- Score ≥ 70 → Eligible for further verification
- Score < 70 → Rejected

🔷 2️⃣ Fraud Detection System

✅ **What it does:**

Detects suspicious or fake applications.

It can identify:

- Duplicate document uploads
- Same income certificate used by multiple students
- Unusual academic patterns
- Fake college data

🔷 3️⃣ Scholarship Recommendation System

✅ **What it does:**

Suggests suitable scholarships to students based on:

- Academic performance
- Income level
- Category
- Location
- Course type

✅ **Example:**

A student with 85% marks and low income
→ System recommends “Merit-cum-Means Scholarship”